

Improved cyclist and helmet detection with hard example mining using YOLOv8

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Introduction

- Bike helmets are a primary safety measure for cyclists, linked to be effective in reducing the risk of injury or fatality in the case of a crash [1].
- Monitoring bike helmet usage is currently done manually, but modern computer vision techniques has shown potential to automate the task.
- Presence of other vehicles, such as scooters and mopeds, can confuse models due to their similarity to bicycles.

Abstract

While interest in measuring bike helmet use has grown, measuring this is primarily a manual task. In automating this task with computer vision, the presence of vehicles resembling bicycles poses a challenge. To address this, we propose hard example mining as a method of reducing false positive predictions, and trackers as a method of facilitating object counting. Our approach has shown to reduce false positives by as much as half for bicycles, and helmets to a lesser extent.



Figure 1. Example image and annotation method.



Figure 2. ROI line applied to frame to facilitate counting.

Materials and Methods

- Dataset is a combination of local GoPro recordings and externally sourced images (Figure 3).
- External data is sourced from the CIMAT cyclist orientation dataset [2].
- Cyclists and helmets are annotated with bounding boxes (Figure 1).
- Models used:
 - YOLOv8
 - BoT-SORT [3]
- Hard example mining [4] is applied to reduce false positive predictions, and a mixed approach that combines hard example mining with the default loss calculation of YOLOv8 is proposed to improve results.
- An ROI line is used to incorporate spatial information when counting objects (Figure 2).

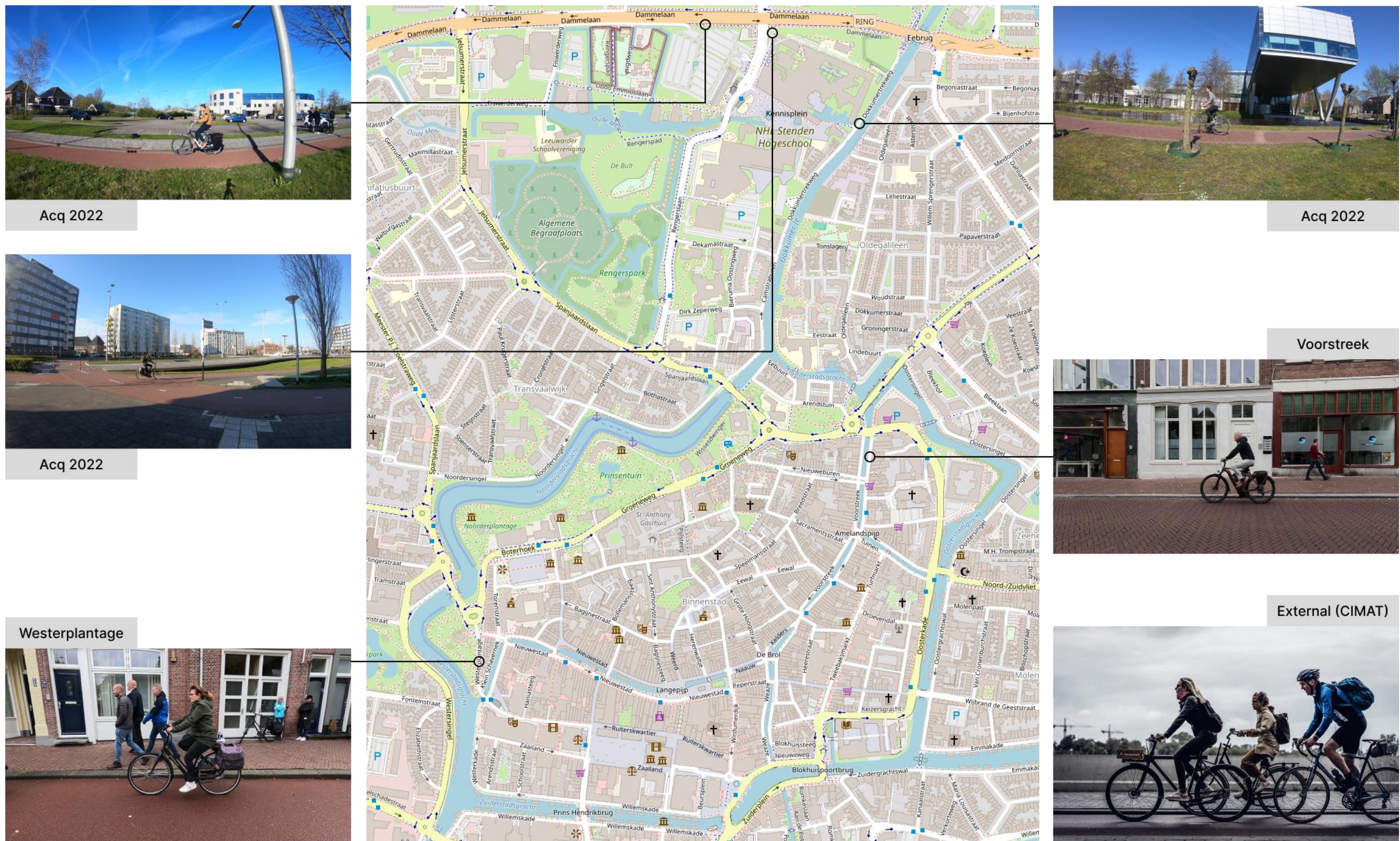


Figure 3. Data collection locations and sample images from each location (© OpenStreetMap contributors).

Acknowledgements

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Experiments and Results

- Models were tested using the default YOLOv8 loss function, the proposed mixed training, and the pure hard example mining loss function, referred to as default, mixed, and mining, respectively in the below results.
- The mixed model showed more stable and reproducible results over five runs, as well as a significantly reduced false positive count for the bicycle class (Figure 4 & Figure 5).

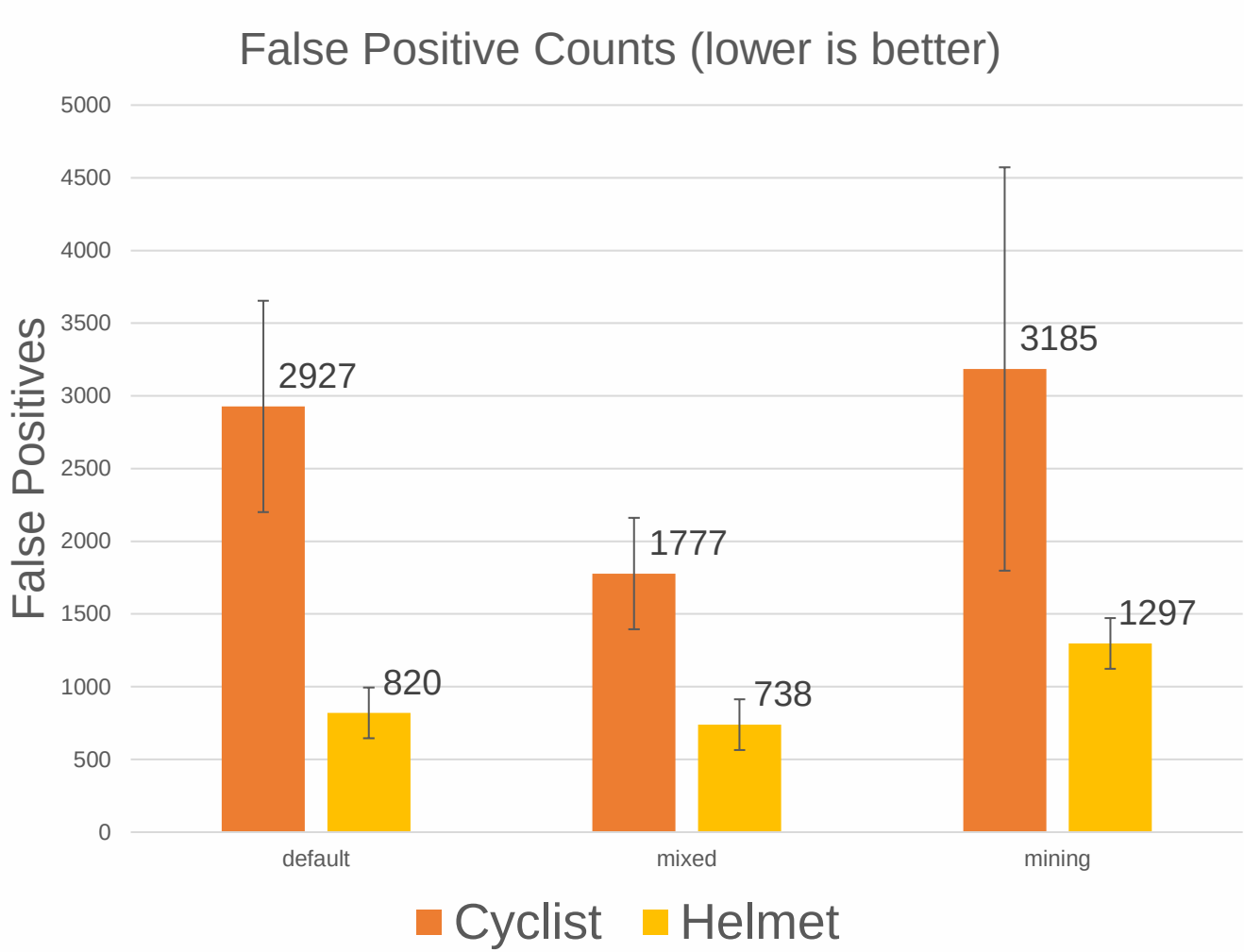


Figure 4. Average false positive counts and standard deviation between the default, mixed, and mining models. Lower numbers represent better results.

Model	Class	MOTA	MOTP
Default	Cyclist	0.58	0.10
	Helmet	0.18	0.30
Mixed	Cyclist	0.66	0.10
	Helmet	0.34	0.30

Table 1. Result comparison between the default and mixed model when using the BoT-SORT tracker.

- Comparison between the default model with the mixed model during tracking showed a higher MOTA when using the mixed model (Table 1).
- Counting results (Table 2) showed an accuracy of 90.3% for counting bicycles, and 34.4% for counting helmets.

Location	Cyclist Accuracy	Helmet Accuracy
Dokkum	81.91%	-
Gytsjerk	90.24%	0.0%
Herbaijum	89.66%	0.0%
Sneek	93.53%	35.29%
Bolsward	87.40%	42.86%
Franeker	84.53%	0.0%
Heerenveen	98.46%	0.0%
Summary	90.34%	34.43%

Table 2. Counting accuracy in different locations and overall counting accuracy based on count confusion matrix. Accuracy values are omitted where no objects were present.



Figure 5a. Cap incorrectly identified as a helmet by the default model.

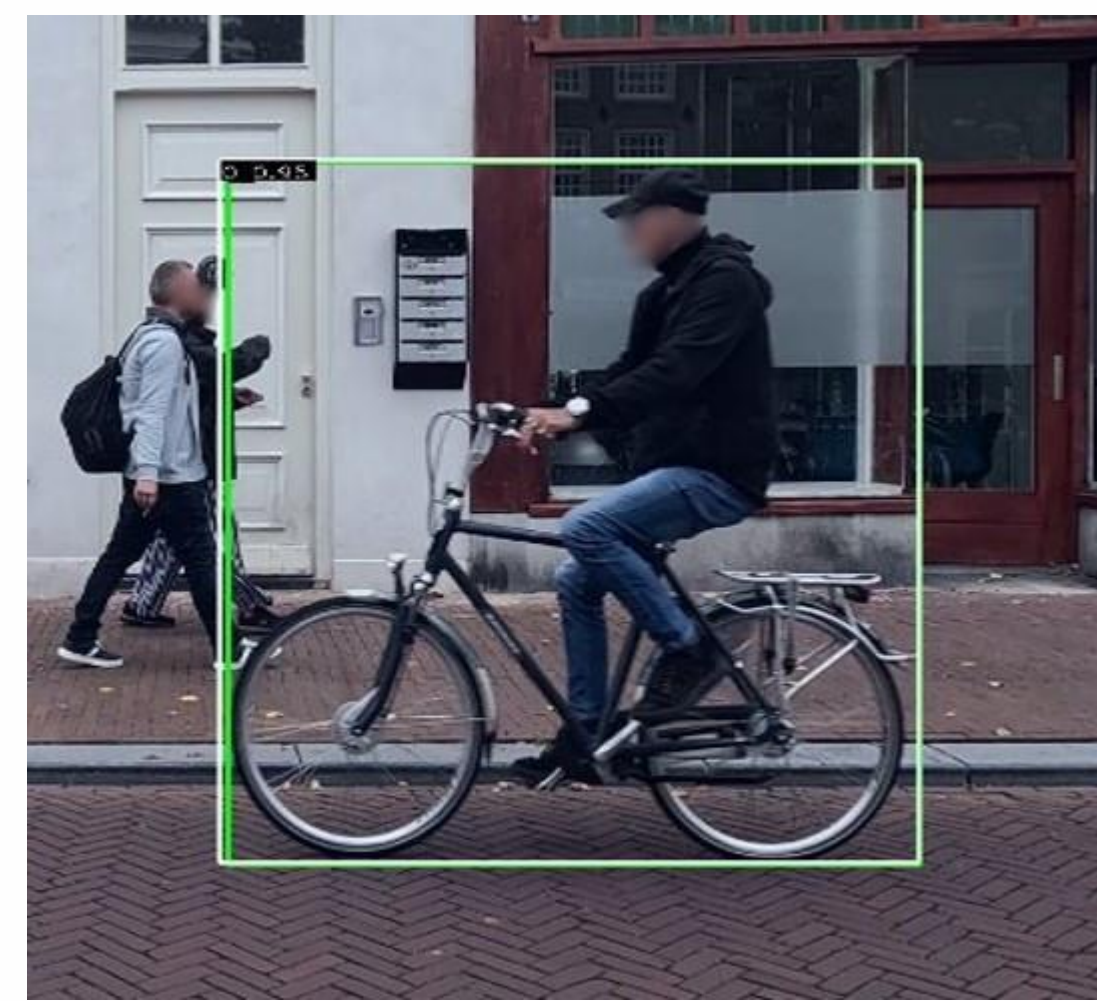


Figure 5b. The same cap as Figure 5a, correctly ignored by the mixed model.

Conclusions

- Using a mixed approach to training that uses both the default YOLOv8 loss and the hard example mining loss has shown to reliably reduce false positive counts.
- The mixed approach to training further improves tracking capabilities when compared to the default YCFX OLOv8 loss.
- Incorporating an ROI line to use spatial information prevents individual false positives detected by the model, but raises issues when objects move through the frame too quickly.

References

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